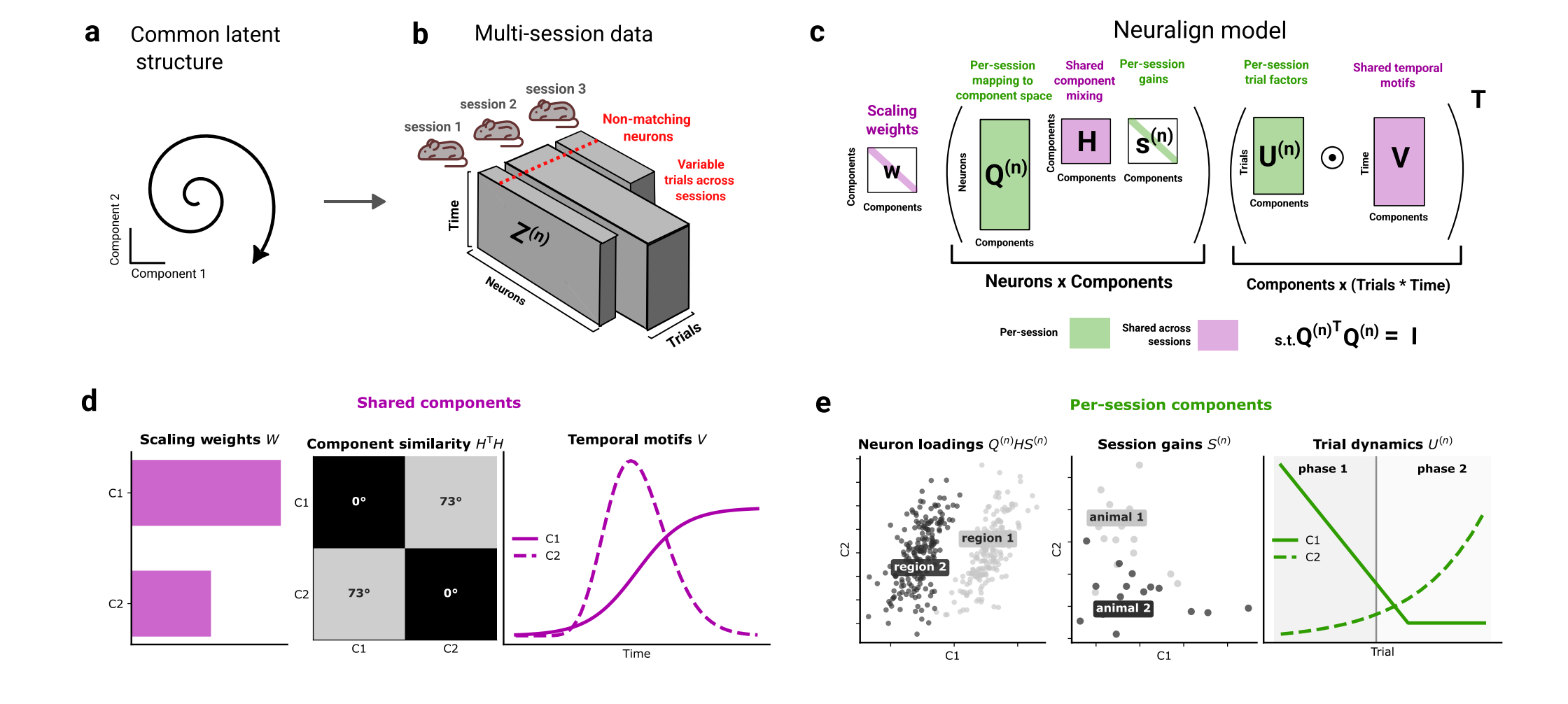

a
Common latent structure
Component 2
Component 1
b
Multi-session data
session 3
session 2
session 1
Non-matching neurons
Variable trials across sessions
Time
Z(n)
Neurons
Trials
c
Neuralign model
Per-session mapping to component space
Shared component mixing
Per-session gains
Per-session trial factors
Shared temporal motifs
Scaling weights
W
Components
Components
Neurons
Q(n)
Components
H
Components
Components
S(n)
Components
T
Trials
U(n)
Components
Time
V
Components
Neurons x Components
Components x (Trials * Time)
Per-session
Shared across sessions
s.t. Q(n)T Q(n) = I
d
Shared components
Scaling weights W
C1
C2
Component similarity HTH
C1
C2
0°
73°
73°
0°
C1
C2
Temporal motifs V
C1
C2
Time
e
Per-session components
Neuron loadings Q(n)HS(n)
C2
region 1
region 2
C1
Session gains S(n)
C2
animal 1
animal 2
C1
Trial dynamics U(n)
phase 1
phase 2
C1
C2
Trial